

ENEL445 Power Systems Optimisation Cheat Sheet

Compact formulas and exam templates

Exam context

Final exam: 2 h, 2 questions, 40 marks. Q2 is mixed Term 2 material: power systems, game theory, LMIs. More mathematical than last year's exam, so prioritise formulas, scaling, and formulations over prose.

Economic dispatch

Meet total demand at minimum cost, ignoring network flows/losses/constraints:

$$\min_{P_{G_i}} \sum_i C_i(P_{G_i}), \quad P_{G_i}^{min} \leq P_{G_i} \leq P_{G_i}^{max}, \quad \sum_i P_{G_i} = P_D$$

Merit order: accept cheapest bids until demand is met. For smooth quadratic dispatch with no active limits:

$$\frac{dC_i}{dP_{G_i}} = \lambda$$

Meaning: active generators have equal marginal cost at optimum.

Penalty factors

With losses:

$$P_D + P_L(P_G) - \sum_i P_{G_i} = 0$$
$$L_i = \frac{1}{1 - \frac{\partial P_L}{\partial P_{G_i}}}, \quad L_i IC_i(P_{G_i}) = \lambda$$

Approximate:

$$\frac{\partial P_L}{\partial P_{G_i}} \approx \frac{\Delta P_L}{\Delta P_{G_i}}$$

Phrase: penalty factor modifies apparent incremental cost to account for how changing generator output changes transmission losses. $L_i > 1$ makes generator appear more expensive; $L_i < 1$ makes it appear cheaper.

Market pricing

Uniform price: NZ and most markets; all accepted generators get the market-clearing price; price is set by the marginal accepted generator; encourages bids closer to cost; low-cost hydro benefits when high-cost gas sets the price.

Pay-as-bid: UK-style example; each accepted generator gets its own bid; payment is set by each bid; bidders try to guess the clearing price; strategic bids can distort dispatch.

Phrase: in uniform pricing, low-cost hydro benefits when expensive gas enters because gas raises the clearing price, and all accepted generators receive that market price. Pay-as-bid seems cheaper, but bidders may inflate bids to guess the clearing price.

ED vs DC OPF vs AC OPF vs SC-OPF

Economic dispatch: total balance + generator costs only. Fast, ignores network.

DC OPF: linearised active-power flows + line limits. Fast enough for real-time dispatch.

AC OPF: full AC power-flow equations, voltage magnitudes/angles, real/reactive power, line/generator/voltage limits. More accurate but large, nonlinear, non-convex.

SC-OPF: OPF plus contingency constraints. Security-aware but much larger.

Why AC OPF is hard

Use all points:

- many variables: $P_G, Q_G, V, \theta, P_{ij}, Q_{ij}$;
- many constraints: power balance, flow equations, voltage/generator/line limits;
- nonlinear $\sin, \cos$ AC equations;
- non-convex equality constraints;
- real-time dispatch needs minutes, not hours;
- SC-OPF repeats/checks constraints under contingencies.

Mitigate: DC OPF surrogate; interior-point solver for full AC OPF; warm start near $V \approx 1$ p.u.; voltage estimates as constants; piecewise linear I^2R losses; AC power-flow feasibility check; contingency screening and heuristic constraints.

Convex/KKT reminder

Convex problem: minimise convex $f_0(x)$ subject to convex inequalities $g_i(x) \leq 0$ and affine equalities $Ax = b$. Nonlinear equality constraints usually destroy convexity.

KKT: stationarity $\nabla_x L = 0$; primal feasibility; dual feasibility $\lambda_i \geq 0$; complementary slackness $\lambda_i g_i(x) = 0$.

Convex + regularity: KKT necessary and sufficient for global optimum. Non-convex: KKT usually only necessary for local optimum.

Algorithm: interior point

Best practical large AC OPF choice. Reason: large smooth nonlinear constrained problem; many inequalities; sparse network structure; scales better than SQP. GA/PSO too slow for huge variable count.

Barrier form:

$$\min_x f(x), \quad g_i(x) \leq 0, \quad h_j(x) = 0$$

$$\min_x f(x) - \mu \sum_i \log(-g_i(x)) \quad s.t. \quad h_j(x) = 0$$

Decrease μ toward constrained optimum.

OPF scaling estimate

Given scaling:

$$O((n + n_g + n_h)^3), \quad N = n + n_g + n_h$$

Then:

$$\frac{t_2}{t_1} \approx \left(\frac{N_2}{N_1} \right)^3$$

With five-fold nonlinear AC slowdown:

$$t_{AC} \approx 5t_{DC} \left(\frac{N_{AC}}{N_{DC}} \right)^3$$

Counting: DC OPF has active powers/angles, active flow constraints, line limits. AC OPF adds voltage magnitudes, reactive powers, nonlinear flow equations, voltage/reactive limits. If told generator count negligible, ignore it.

DC OPF

DC OPF is a DC approximation of AC power flow, not a DC grid model.

Assumptions:

- ignore reactive power;
- $V_i = 1$ p.u.;
- small angle differences:

$$\sin(\theta_i - \theta_j) \approx \theta_i - \theta_j, \quad \cos(\theta_i - \theta_j) \approx 1$$

- negligible resistance, lossless network.

Linear active-power flow:

$$P_{ij} = B_{ij}(\theta_i - \theta_j)$$

Industry workflow: solve DC OPF; estimate losses if needed using approximate current and piecewise linear I^2R ; run standard AC power flow to check feasibility; add heuristic/security constraints and re-solve if infeasible.

Important: AC power-flow check after DC OPF is not AC OPF.

SC-OPF

Definition: dispatch remains feasible under credible contingencies, e.g. line/generator outages.

Difference: regular OPF checks base case; SC-OPF checks base case plus contingency cases.

Why hard: each contingency adds/checks another set of operational constraints; many contingencies possible; can be orders of magnitude larger than AC OPF.

Practice: solve DC OPF, check selected contingencies, add heuristic constraints, re-solve until acceptable.

State estimation

Estimate voltage states from noisy/incomplete measurements:

$$z = h(x) + e$$

Weighted least squares:

$$\min_x (z - h(x))^T R^{-1} (z - h(x))$$

Equal standard deviations:

$$\min_x \|z - h(x)\|_2^2$$

Linear case $h(x) = Hx$:

$$\min_x \|z - Hx\|_2^2, \quad \hat{x} = (H^T H)^{-1} H^T z$$

Weighted:

$$\hat{x} = (H^T R^{-1} H)^{-1} H^T R^{-1} z$$

If only angle differences are observable, choose a reference angle.

Cases: under-determined = not enough measurements, usually add pseudo-measurements; over-determined = more measurements than states, usual least-squares case. $R = \text{diag}(\sigma_i^2)$; large σ_i means low trust/low weight. Per-unit conversion: $P_{\text{p.u.}} = P_{\text{actual}}/S_{\text{base}}$, so 1 MW/100 MW = 0.01 p.u.

Linear SE template

Given:

$$P_{ij} = B_{ij}(\theta_i - \theta_j), \quad z = [P_{12}^m, P_{13}^m, P_{32}^m]^T$$

Set $\theta_1 = 0$, $x = [\theta_2, \theta_3]^T$:

$$H = \begin{bmatrix} -B_{12} & 0 \\ 0 & -B_{13} \\ -B_{32} & B_{32} \end{bmatrix}$$

because $P_{12} = B_{12}(0 - \theta_2)$, $P_{13} = B_{13}(0 - \theta_3)$, $P_{32} = B_{32}(\theta_3 - \theta_2)$.

Answer:

$$\min_x \|z - Hx\|_2^2$$

Define z, x, H .

Nonlinear SE template

AC measurement functions:

$$P_{ij}(x) = V_i V_j B_{ij} \sin(\theta_i - \theta_j)$$

$$Q_{ij}(x) = V_i (V_i - V_j \cos(\theta_i - \theta_j)) B_{ij}$$

Example:

$$x = [V_1, V_2, V_3, \theta_1, \theta_2, \theta_3]^T$$

If unknown parameter, include it, e.g. B_{32} :

$$x = [V_1, V_2, V_3, \theta_1, \theta_2, \theta_3, B_{32}]^T$$

Build z from measured P, Q, V, θ . Build $h(x)$ with corresponding predicted measurements, then write:

$$\min_x \|z - h(x)\|_2^2$$

Do not solve unless asked.

Short phrases

- ED ignores network; OPF includes network physics.
- AC OPF is nonlinear/non-convex due to AC flow equations.
- DC OPF is a fast linear surrogate.
- SC-OPF adds contingency constraints.
- LMPs come from OPF Lagrange multipliers.
- Pseudo-measurements have large standard deviation and low weight.
- Define variables before writing the objective.